Politecnico di Torino

Lab report for the course Qubit Electronics

Master degree in Quantum Engineering

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1. Introduction

The objective of this laboratory is to study the electrostatic behaviour of a silicon MOS double quantum dot through a numerical simulation workflow. Starting from the device geometry, the report analyzes the structure of the mesh, the main material regions, and the role of the different electrostatic gates used to control the confinement potential.

The simulation workflow uses Gmsh to define and inspect the device geometry and its mesh, while the non-linear Poisson equation is solved with gate boundary conditions corresponding to the reservoir, barrier, plunger, and lateral confinement electrodes. The resulting conduction-band energy profile is then visualized in ParaView through slices and line plots, allowing the formation and coupling of the two quantum dots to be analyzed.

2. Mesh

2.1 Structure Analysis

The device analyzed in this laboratory is a Double Quantum Dot structure fabricated using Silicon MOS technology. The overall device has the dimensions as reported in the table 2.1.

Axis	Length (nm)
X	353
Y	273
Z	126

Table 2.1: Device Dimensions

The device is divided into different regions determined by the code itself, in order to assign different properties both in terms of phyical caracteristics and in terms of meshing for the calculation of the simulation. Regarding the mesh sizes, in the code are defined different measures that will be assigned to specific regions:

```
grid_spacing_01          = 1; //nm
grid_spacing_02          = grid_spacing_01*2;
grid_spacing_03          = grid_spacing_02*2;
grid_spacing_04          = grid_spacing_03;
grid_spacing_05          = grid_spacing_03*2;
```

Each of this parameter will indicates the local characteristic mesh size assigned to the points of the regions that will constitute the device. Of course, the higher the mesh size, the higher the elements size and so the lower of the mesh density, resulting in a lower accuracy when it comes to perform calcualtions.
Each point is defined as:

```
Point(id) = {x, y, z, lc};
```

where the last arguement is the one just mentioned.
By analyzing the code, it is possible to indentify the division of the structure, which is shown in table 2.2 starting from the bottom to the top, with their respective mesh sizes.

Region name	Mesh size (nm)
region_bulk_bottom	8
region_pre_quantum_A/B/C (bottom face)	2
region_pre_quantum_A/B/C (top face)	1
region_quantum_A/B/C	1
region_post_quantum_A/B/C	2
region_bulk_top	2
region_bottom_oxide_layer	4
region_Y_gate	4
region_top_oxide_layer	8
region_Barrier/Plunger/Reservoir_X_gate	4

Table 2.2: Device regions and mesh sizes

As we can see, the regions with the highest mesh density are the ones that correponds to the volume of space where the quantum dots are located, togheter with the regions right above and right below them. This is obvious, since here we need to calculate the poisson equation with the highest accuracy.

In order to have a better understanding of the overall structure, we can observe the graphical representation in figure 2.1 of the .geo file, in particular visualizing the 2D meshes, since with the 3D mesh is difficult to distinguish the different components.

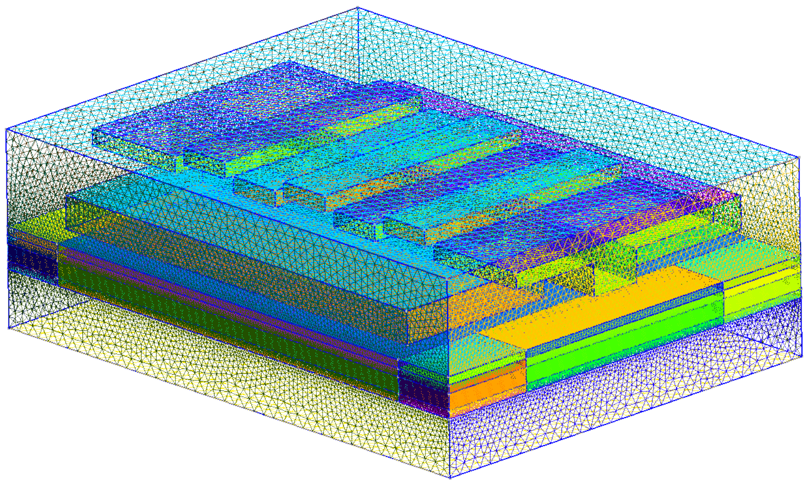

Figure 2.1: 2D mesh of the device

Figure 2.1 illustrates the two-dimensional mesh of the double quantum dot structure. As previously described, the mesh density varies across the different regions to optimize the computational accuracy and simulation time. Each functional region is clearly distinguishable due to a specific color. It is worth to analyze the layout of the gates used in this type of double quantum dot structure: positioned directly above the active region, we find a pair of 'Y-gates' located at both ends of the y-axis (the short side). Their primary function is to provide lateral confinement, narrowing the silicon channel at the center of the device to create a quasi-onedimensional path for electrons. Above a thin dielectric layer, the 'X-gates' are visible. This layer includes three barrier gates—which define the tunnel junctions between the dots and the reservoirs, as well as the inter-dot barrier—and two plunger gates used to precisely tune the energy levels within each quantum dot. Additionally, two gates are positioned above the reservoir regions to regulate electron density and chemical potential. A key feature of the X-gates is their characteristic T-shape. This geometry is designed to maximize the electrostatic coupling with the silicon channel. By partially wrapping around the sides of the channel, the T-shaped gates provide a more effective pinch-off compared to a simple planar gate. This ensures that the potential barriers are sharp and well-defined, which is critical for isolating individual electrons in the Coulomb blockade regime.

3. Poisson

3.1 Simulation setup

The Poisson stage starts from the three-dimensional SiMOS mesh, which is converted into a QTCAD device configured for electron confinement:

```
d = Device(mesh, conf_carriers="e")
d.set_temperature(T)
d.statistics = "FD_approx"
```

The option `conf_carriers="e"` tells the solver that the mobile charge density to be computed is electronic. The temperature is set to 15 mK, consistent with the cryogenic operating regime of the double quantum dot, and the Fermi–Dirac approximation is used to describe the occupation of the available electronic states during the self-consistent Poisson calculation:

$$T = 0.015\mathrm{K} = 15\mathrm{mK}.$$

The electrostatic potential is controlled by the gate surfaces defined in the geometry rather than by assigning arbitrary voltages inside the device volume. Each surface declared with `new_gate_bnd` creates a boundary condition for one metallic gate, and all gates share the same work-function offset,

$$\Phi_m = \chi_{\mathrm{Si}} + \frac{E_{g,\mathrm{Si}}}{2},$$

corresponding to a mid-gap silicon reference. The different voltages in Table 3.1 then represent the intentional tuning knobs of the device: reservoir gates accumulate carriers, barrier gates control tunnel coupling, and plunger gates tune the energy levels of the two dots.

Boundary	Voltage (V)	Role
`surface_Y1_gate`	0.535	Longitudinal confinement gate.
`surface_Y2_gate`	0.535	Longitudinal confinement gate, biased symmetrically with Y1.
`surface_Reservoir_L_X_gate`	1.300	Left reservoir accumulation.
`surface_Reservoir_R_X_gate`	1.300	Right reservoir accumulation.
`surface_Barrier_L_X_gate`	0.040	Left tunnel barrier.
`surface_Barrier_M_X_gate`	0.400	Inter-dot barrier.
`surface_Barrier_R_X_gate`	0.040	Right tunnel barrier.
`surface_Plunger_L_X_gate`	0.700	Left dot plunger.
`surface_Plunger_R_X_gate`	0.7001990578	Right dot plunger, detuned by about 199 µV.

Table 3.1: Gate boundary voltages used in the Poisson simulation.

Therefore, the simulation does not fix the carrier population through explicit source or drain ohmic contacts. The reservoirs are still present in the mesh and in the material definition, but their role is imposed indirectly through the high reservoir-gate voltages. This means that the formation of the accumulated regions and their coupling to the dot are determined by the electrostatic solution rather than by a separate ohmic-contact boundary condition.

3.2 Materials and regions

Before the boundary conditions are applied, each mesh region is assigned a material. This assignment fixes the dielectric response used by the Poisson equation; in silicon it also provides the band parameters needed to compute the electron density. The three QTCAD material objects used in the device are reported in Table 3.2.

QTCAD material	Physical material	Use in the device
`mt.Si`	Silicon	Semiconductor body, reservoirs, and quantum regions. Some silicon regions are n-doped with $N_D = 5\times10^{24}\mathrm{m}^{-3}$, corresponding to $5\times10^{18}\mathrm{cm}^{-3}$; the p-doping is set to zero.
`mt.SiO2`	Silicon dioxide	Oxide regions immediately above the quantum layer and in the upper central stack. These regions act as insulating dielectrics and do not contribute mobile carriers.
`mt.Al2O3`	Aluminum oxide	Top oxide and gate-stack regions. The electrostatic action of the gates is imposed separately through the gate boundary conditions.

Table 3.2: Materials appearing in `SiMOS_poisson.py`.

The same region labels are then used to identify the central quantum-dot stack:

`dot_region` = {`region_pre_quantum_B_mid`,`region_quantum_B_mid`,`region_post_quantum_B_mid`}.

The definition of `dot_region` is therefore a selection of the central quantum-dot volume, not an additional material assignment. By excluding this region from the classical charge-density calculation, QTCAD avoids treating the dot as a continuous electron gas with strong screening. This is important because the central region is expected to host only a small number of confined electrons, so its electrostatic behavior must remain compatible with a later quantum treatment instead of being fully classical.

3.3 Poisson solver parameters

The solver is instantiated with a geometry file,

```
p = PoissonSolver(d, solver_params=params_poisson, geo_file=path_geo)
```

so QTCAD uses the adaptive-mesh non-linear Poisson solver. The meaning of each parameter in the script is summarized in Table 3.3.

Parameter	Value	Meaning in this simulation
`tol`	10^{-7}	Convergence tolerance for the self-consistent Poisson loop. With the default absolute error criterion, the iteration stops when the maximum change in electrostatic potential between successive iterations is below 10^{-7}V.
`initial_ref_factor`	0.55	Initial adaptive-refinement factor. It is dimensionless; values closer to zero force stronger mesh refinement after non-convergence, while values closer to one refine more conservatively.
`final_ref_factor`	0.85	Refinement factor used once the solver error has saturated. The high value means the final adaptive refinements are intentionally mild, avoiding an unnecessarily large mesh.
`min_nodes`	0	No lower bound is imposed on the total number of mesh nodes. If the electrostatic solution converges, the adaptive procedure is allowed to stop even for a relatively small mesh.
`maxiter`	1000	Maximum number of self-consistent non-linear Poisson iterations before the solve is considered unsuccessful. The high value leaves room for convergence at the low operating temperature used in the simulation.
`refined_region`	`dot_region`	Region in which QTCAD enforces a maximum local mesh characteristic length. Here the enforced refinement is focused on the central dot stack.
`h_refined`	0.3	Maximum characteristic mesh length in the regions listed in `refined_region`. Since the mesh is loaded with a 10^{-9} scaling factor and the Gmsh geometry is in nanometers, this corresponds to 0.3nm.
`max_nodes`	10^{10}	Upper bound on the number of mesh nodes allowed during adaptive refinement. This value is effectively non-limiting for the present calculation; the practical bound will be memory and runtime.

Table 3.3: Meaning of the Poisson solver parameters used in the adaptive calculation.

Taken together, these settings make the solution accurate where the confinement potential is most sensitive, without forcing the whole mesh to the same resolution. The small tolerance controls the self-consistent potential, while `h_refined` concentrates the finest elements in `dot_region`. Away from the central stack, the relatively mild refinement factors allow the adaptive solver to keep a coarser mesh where the potential varies more smoothly.

4. Quantum dot characterization with Paraview

To analyze the charge carrier confinement within the device, the simulation results were processed using Paraview. The following sections describe the charge density distribution and the resulting potential landscape.

4.1 Conduction band energy distribution

The conduction band energy distribution was visualized by performing a *Slice* operation on the 3D model. The slice was taken with a normal vector along the **Z-direction** at the coordinate $\mathbf{z = 45}$, which corresponds to the active region of the SiMOS structure where the dots are formed.

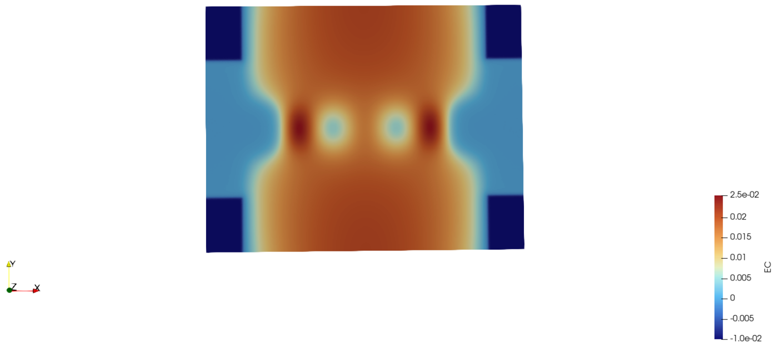

Figure 4.1: Conduction band energy map E_c showing the electrostatic confinement potential for the double quantum dot.

The map in Figure 4.1 illustrates the spatial arrangement of the potential energy levels. The regions characterized by the lowest energy are represented in dark blue, which corresponds to the potential wells of the reservoirs and the interior of the dots where electrons accumulate. Conversely, the areas in intense red represent high energy levels, indicating the presence of electrostatic potential barriers that provide the necessary confinement. In the central area, two distinct potential minima are visible, identifying the double quantum dot system.

4.2 Potential profile analysis

To further investigate the confinement, a *Plot Over Line* was performed across the domain along the x-axis, at the fixed coordinates $y = 136.5$ and $z = 45$. This profile provides a quantitative view of the potential landscape through the centers of the two dots.

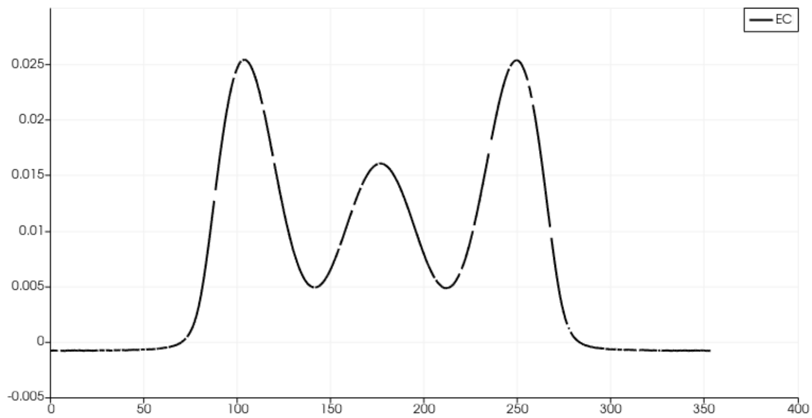

Figure 4.2: Potential graph (EC) along the x-axis.

The resulting graph in Figure 4.2 confirms the formation of the double quantum dot. However, contrary to our previous laboratory observations, the central potential barrier between the two dots appears significantly lower than, or comparable to, the outer confinement barriers separating the dots from the reservoirs. This behavior, resulting from the specific gate bias configuration applied in the SiMOS device simulation, suggests a different coupling regime between the two quantum dots in this specific instance